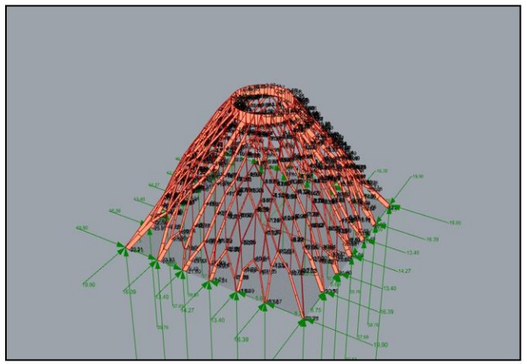
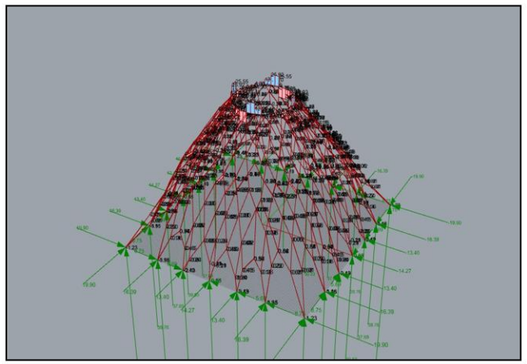
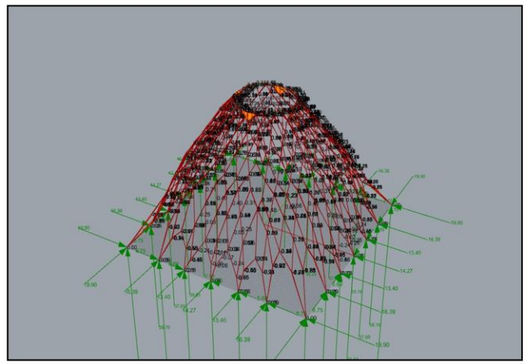
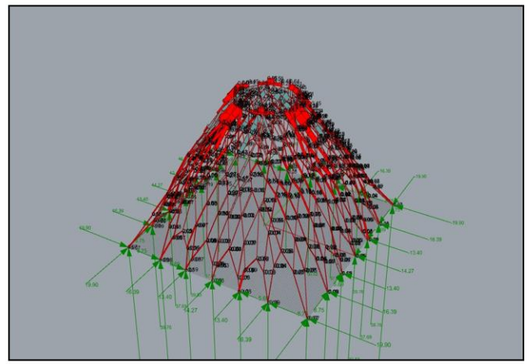
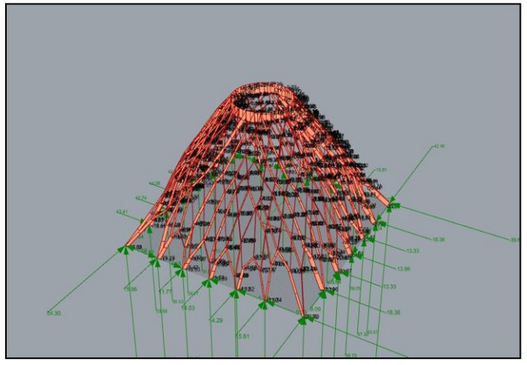
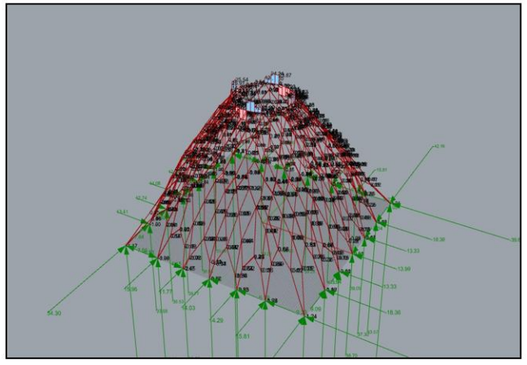
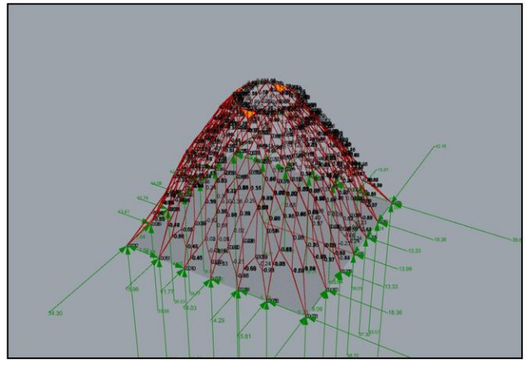
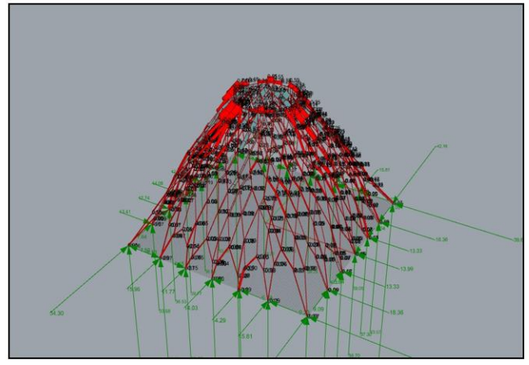
05 THE MODEL UNDER LOAD: FORCES AND MOMENTS TOGETHER - ITERATION 19

One run of the whole PolyFrame net (19: gradual branching, 24 supports, 3 m crown ring) under LONG (weight) and SHORT (weight + typhoon), drawn the same way for every force: same camera, same scale for LONG and SHORT, values on the bands in kN and kNm. Ribs 150 x 250 mm, cast C30/37.

	AXIAL FORCE Nx	SHEAR Vz	BENDING My	TORSION Mx
LONG	 crown ring -51 kN (compression); net -1.5 to -29 kN; no bar in tension	 26 kN in the crown ring; under 5 kN in the net	 10.5 kNm in the crown ring; under 2.1 kNm in the net	 0.6 kNm at most
SHORT	 crown ring down to -54 kN; two upper bars +2.3 kN (tension)	 26 kN in the crown ring; under 5 kN in the net	 10.7 kNm in the crown ring; still under 2.1 kNm in the net	 0.7 kNm at most

BEAM FORCES FROM KARAMBA, MIN / MAX OF EACH MEMBER GROUP (kN, kNm; + tension)

	LONG			SHORT		
	N	My	Mz	N	My	Mz
Foot bars (48)	-23.9 / -18.6	-1.3 / +0.3	+-0.4	-27.8 / -13.7	-1.4 / +0.4	+-0.5
Lower net, to 2.5 m (120)	-13.9 / -4.0	-0.9 / +0.5	+-0.5	-16.0 / -3.1	-1.1 / +0.6	+-0.6
Middle net, 2.5-4.5 m (208)	-16.2 / -2.0	-0.6 / +1.9	+-1.1	-18.4 / -1.4	-0.7 / +1.9	+-1.1
Upper net, 4.5-6.67 m (168)	-28.6 / -1.5	-2.1 / +1.3	+-0.8	-32.1 / +2.3	-2.1 / +1.3	+-0.8
Crown ring (24)	-51.1 / -49.2	-10.5 / +3.3	-3.3 / +1.9	-54.1 / -40.2	-10.7 / +3.6	-4.0 / +2.1

Global check: LONG SUM V = 1691.6 kN = the weight (ribs 525 + casing 1167). SHORT SUM V = 1607 kN (the typhoon lifts 85 kN) and SUM H = 113 kN, shared by the 24 supports.

HOW TO READ THE DIAGRAMS

Each band is drawn on its member; its width is the force there and the numbers give it in kN (Nx, Vz) or kNm (My, Mx). The two colours are the two signs.

Nx: + tension, - compression. My bends a member in its depth; Mx twists it.

Karamba scale, m per kN or kNm: Nx 0.01, Vz 0.02, My 0.04, Mx 0.6, so torsion looks big but is small.

SHORT: 1.0 G + 1.0 W, qp 2.6 kN/m2 (V0 46 m/s, assumed), wind toward +X, cpe +0.7 / -0.6 / -0.7 on the casing faces. Green arrows: support reactions (kN).